Unit 4: Sensors

Color Sensor

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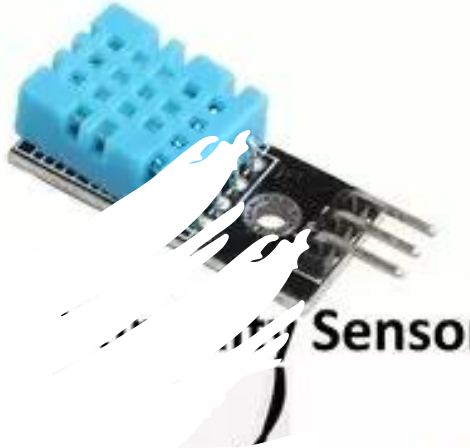
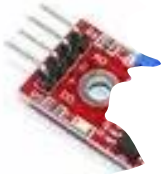


Sensor

Air Pressure Sensor

Gyro + Accelero Sensor



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Infrared Sensor

IR Receiver

Temp. Sensor

Pulse Oximeter

Gas Sensor

Light Sensors

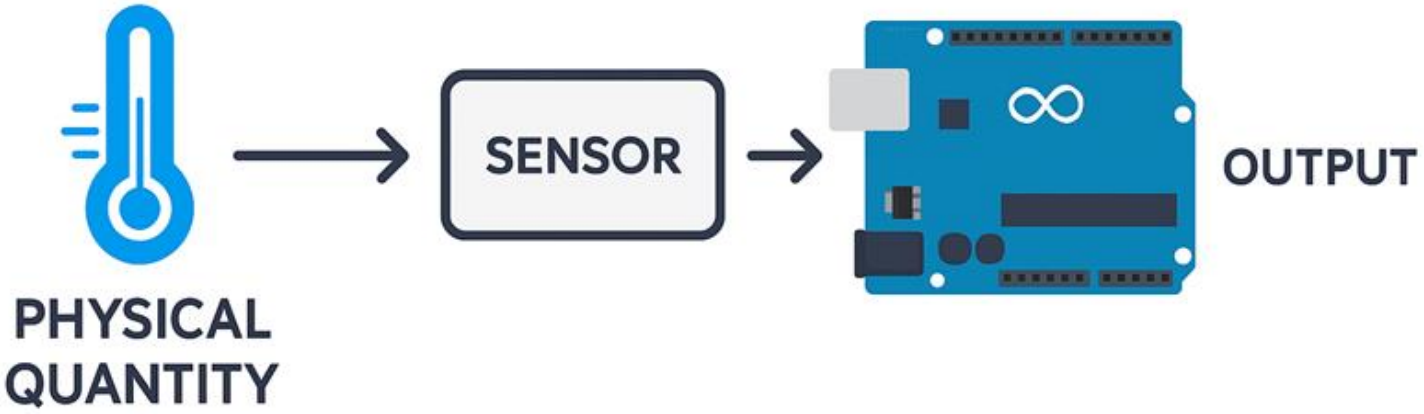
Load Sensor

What are Sensors?

A sensor is an electronic component or device that detects physical quantities (temperature, light, distance, pressure, etc.) and converts them into electrical signals.

This is then sent to a display or transmitted, or relayed to other electronic devices for further processing.

Sensors are classified into various types based on their functionality and the physical phenomena they measure.



General Working Principle:

Sensing (Detection):
The sensor detects changes in the physical environment (e.g., temperature rise, light intensity, object motion).

Conversion (Transduction):
The detected physical quantity is converted into an electrical signal by a transducer element inside the sensor.

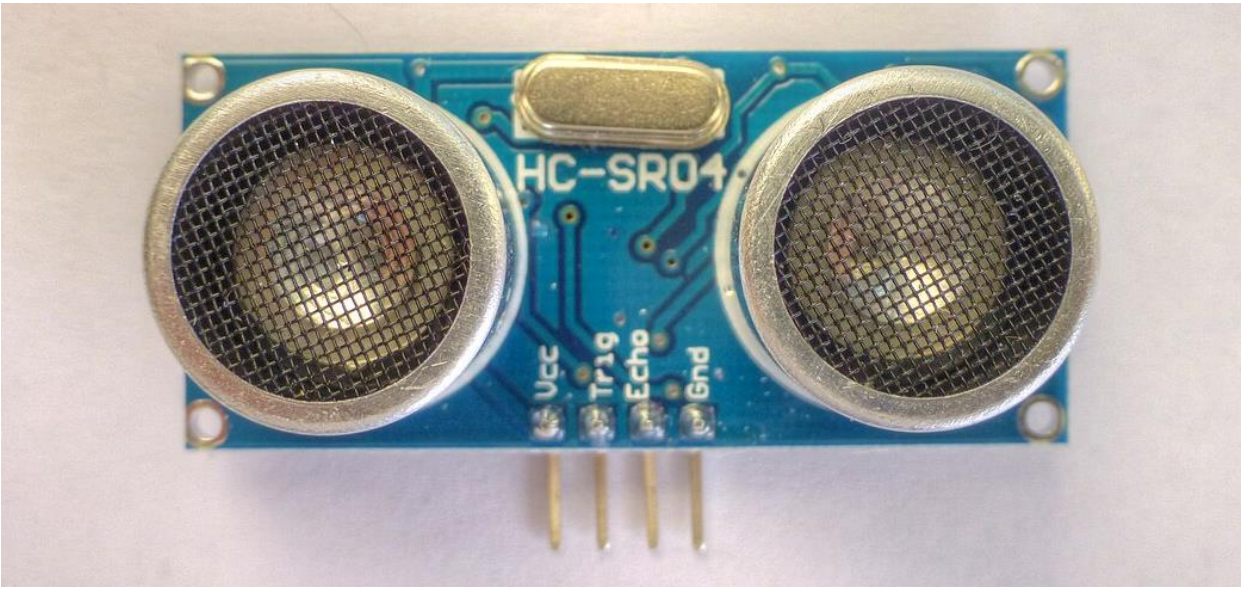
Example: *LDR: Light changes its resistance.*
LM35: Temperature changes its output voltage.

Signal Conditioning (Optional):
Some sensors include amplifiers or filters to strengthen or stabilize the signal.

Output to Microcontroller:
Analog Sensors: *Provide a continuous varying voltage (0–5V on Arduino).*
Digital Sensors*: Provide discrete signals – either HIGH (1) or LOW (0).*

Ultrasonic Sensor

An ultrasonic sensor is an instrument that measures the distance to an object using ultrasonic sound waves.



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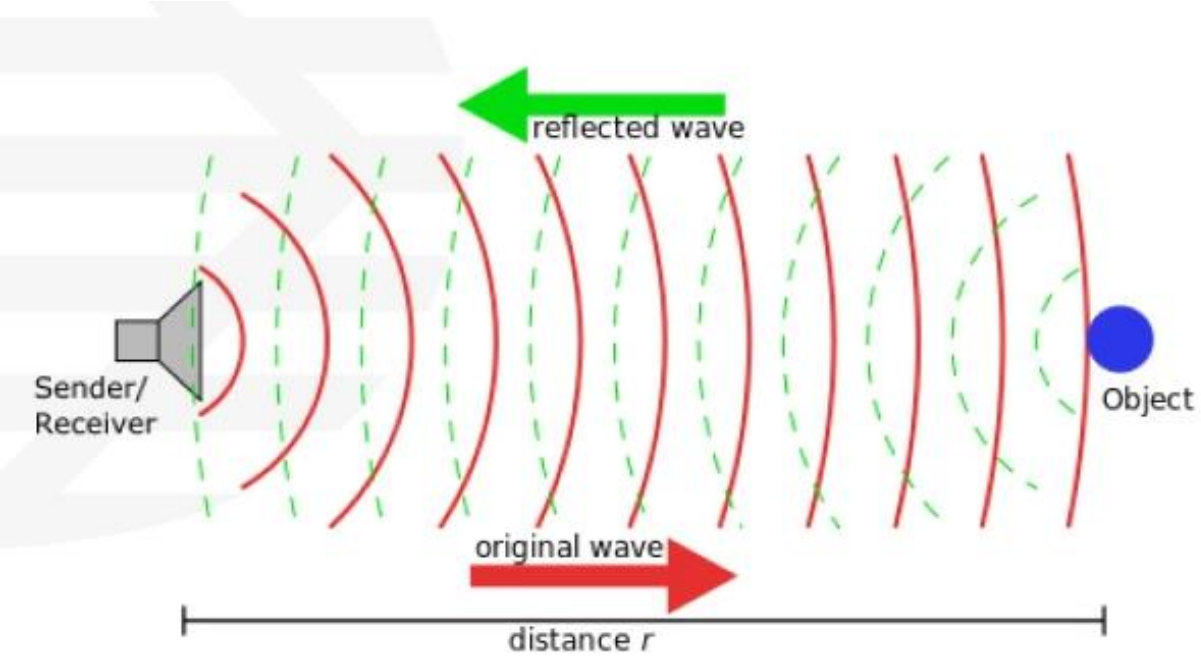
The sensor consists of two primary components: a transmitter and a receiver .

The transmitter is responsible for emitting a high-frequency sound. In essence, ultrasonic refers to frequencies beyond the range of the human hearing - so something higher than 20kHz.

When the sound wave hits an object, it bounces back. This reflected wave is detected by the receiver. The sensor will use the micro-controller (Arduino) internal clock to find out how much time it took for the sound to bounce back. This small clock turns on when a high-frequency wave is emitted and turns off when its reflection is detected.



Applications of Ultrasonic Sensor (HC-SR04)

- **Distance Measurement**
  Used in robotics and automation to measure distance between the sensor and an object.

- **Obstacle Avoidance in Robots**
  Detects obstacles in autonomous robots, line-following robots, and drones.

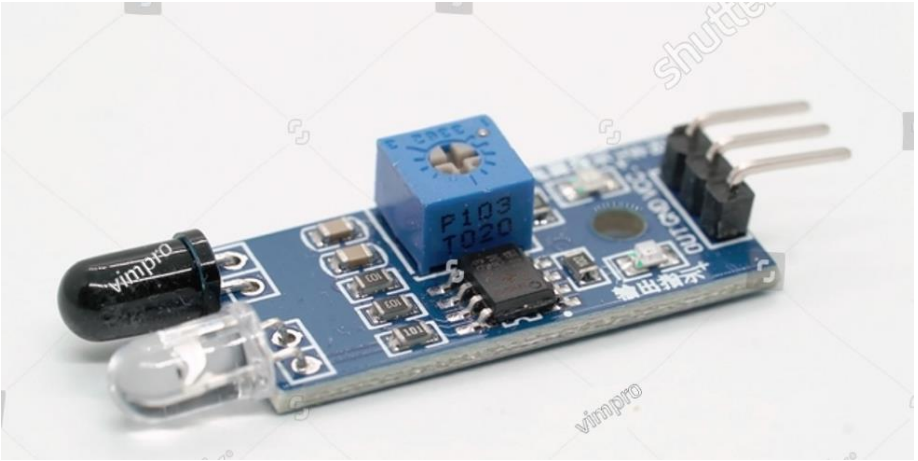
- **Level Detection**
  Measures liquid levels in tanks (water, oil) without direct contact.

- **Parking Assistance Systems**
  Used in vehicles to detect objects during parking (reverse parking sensors).

- **Security Systems**
  Intruder detection by sensing movement within a defined range.

Active IR Sensor

An Infrared (IR) Sensor is an electronic device that emits and detects infrared radiation to sense objects, distance, or motion.



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Working Principle:

- Emits infrared light using an IR LED.
- The reflected IR light from an object is detected by a photodiode or phototransistor.
- The sensor converts the detected light into an electrical signal.

Applications

- **Object Detection**

  Detect objects in front of a robot to avoid collisions or change its path.

- **Line Following Robots**

  Sense the difference in light reflection between black and white surfaces to help robots follow a line or track.

- **Proximity Sensors**

  Detect when a person or object comes close, triggering automatic door opening or counting items on a conveyor.

- **Intruder Alarms**

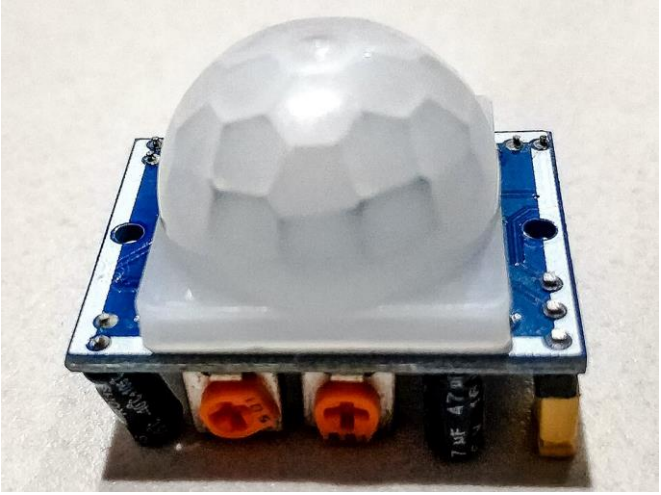
  Triggers an alarm when an IR beam is broken by an intruder.

Passive IR Sensor

A **Passive Infrared (PIR) sensor** detects motion by sensing infrared radiation (heat) naturally emitted by humans or animals.

Working Principle:

- The sensor has **pyroelectric elements** that detect changes in infrared radiation levels.
- When a warm body (like a human) moves in front of it, the change in IR levels triggers a **digital HIGH output**.



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Applications:

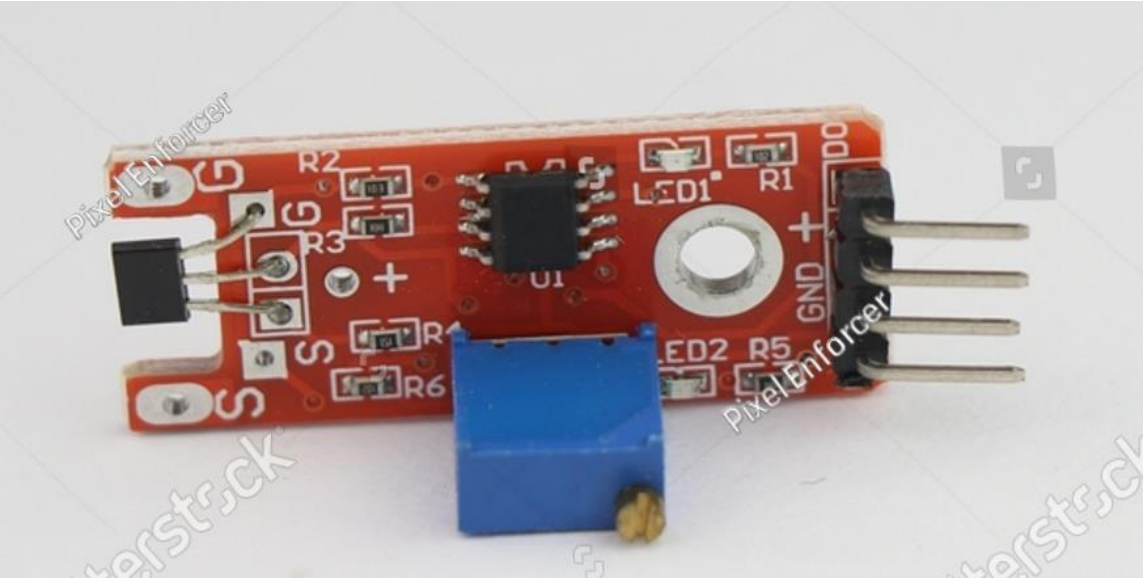
- **Security Alarms and Automatic lights.**
  Senses movement by detecting changes in infrared radiation (body heat). When motion is detected, it triggers alarms for security or turns on lights automatically

- **Smart Home Automation:**
  Controls appliances like fans or air conditioners based on human presence.

- **Robotics:**
  Detect humans or animals nearby, enabling interactive responses or avoiding collisions.

Linear Magnetic Hall Sensor

A Linear Hall Sensor is a sensor that measures the strength of a magnetic field and converts it into a proportional analog voltage output.



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Working Principle

- Based on the Hall Effect:
  When a current-carrying conductor or semiconductor is placed in a magnetic field, a voltage (Hall voltage) is generated perpendicular to both the current and the magnetic field.

- The sensor outputs a continuous analog voltage proportional to the magnetic field strength.

Applications:

- **Magnetic field measurement**
  Measure the strength and polarity of magnetic fields for educational or research purposes.

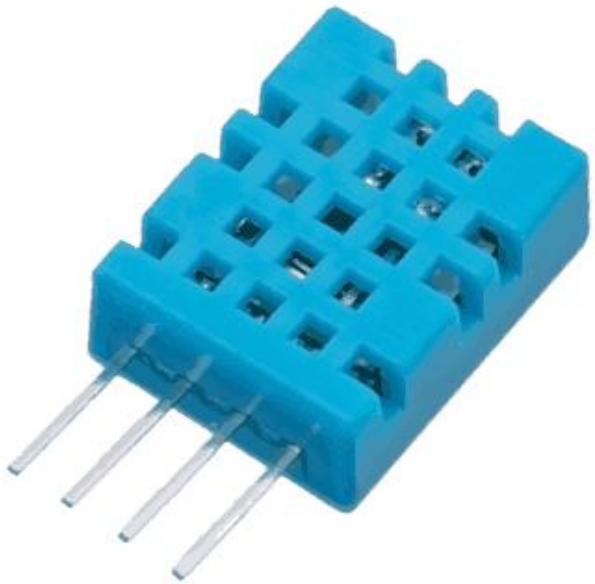
- **Rotary position sensing**
  Detects the angular position of a rotating magnet attached to a motor shaft, useful for motor control systems.

- **Speed measurement**
  Measures rotation speed (RPM) by counting how many times a magnet passes by the sensor per second.

- **Proximity detection**
  Detects when a magnet comes close, used in door sensors, safety switches, or contactless triggers.

DTH 11

DHT11 means it is a **Digital Humidity and Temperature sensor** (model 11)
The **DHT11** is a **digital sensor** that measures **temperature** and **relative humidity**. It is low-cost and easy to interface with Arduino.



Working Principle:

- The sensor has:
  - A thermistor to measure temperature.
  - A capacitive humidity sensor to measure moisture in the air.

- An internal microcontroller processes these readings and sends them as a digital signal to the Arduino.

Applications:

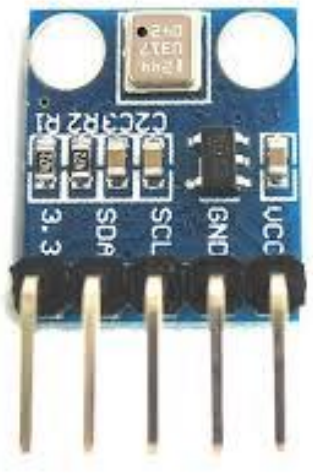
- **Weather monitoring systems**
  Used to track temperature and humidity in real-time for weather prediction and recording environmental conditions

- **Smart home automation**
  *Automatically controls **fans, air conditioners, or dehumidifiers** based on room temperature and humidity levels.*

- **Greenhouse monitoring**
  Maintains optimal temperature and humidity for healthy plant growth by triggering misting systems or ventilation.

- **HVAC systems**
  Ensures efficient climate control by adjusting heating or cooling equipment based on sensor readings.

Barometer Sensor (BMP180)



The BMP180 is a digital barometric pressure sensor that measures atmospheric pressure and can also estimate altitude and temperature.

Working Principle:

- Works on the **piezo-resistive principle**:
  - A thin silicon diaphragm with piezo-resistive elements bends under atmospheric pressure.
  - This bending changes electrical resistance, which is converted to a digital pressure value by an internal ADC and microcontroller.

- Using pressure readings, altitude is calculated based on standard atmospheric pressure at sea level.

Applications:

- **Weather stations**
  Measures atmospheric pressure to help predict weather changes (e.g., falling pressure indicates rain, rising pressure indicates clear weather).

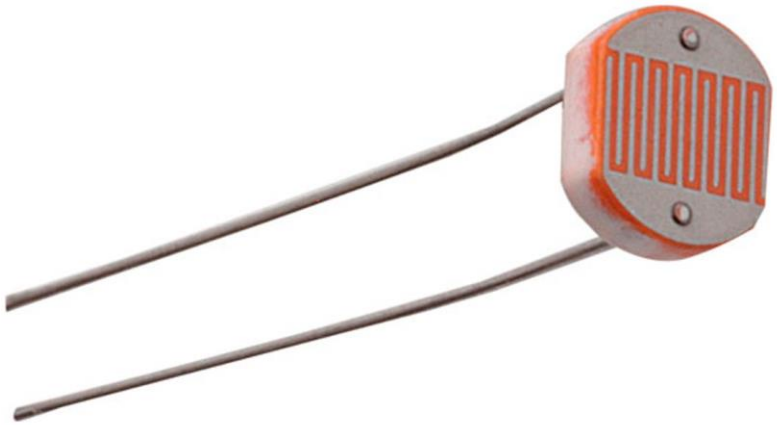
- **Altimeters**
  Calculates altitude based on air pressure, useful in drones, airplanes, and GPS devices for navigation and height tracking.

- **Indoor Navigation**
  Detects small pressure changes to determine floor level changes inside multi-storey buildings (useful in smart navigation systems).

- **Temperature monitoring**
  Built-in temperature sensor provides environmental temperature readings, useful for climate control and data logging.

LDR

An **LDR (Light Dependent Resistor)** is a **passive light sensor** whose resistance changes with the amount of light falling on it.



Working Principle:

- High Light Intensity: Resistance decreases, allowing more current to flow.
- Low Light Intensity (Darkness): Resistance increases, reducing current flow.
- Arduino reads the voltage change across the LDR using **analogRead().**

Applications

- **Automatic Street Lights:**
  Turns ON lights when it's dark (high resistance) and OFF during daylight (low resistance).

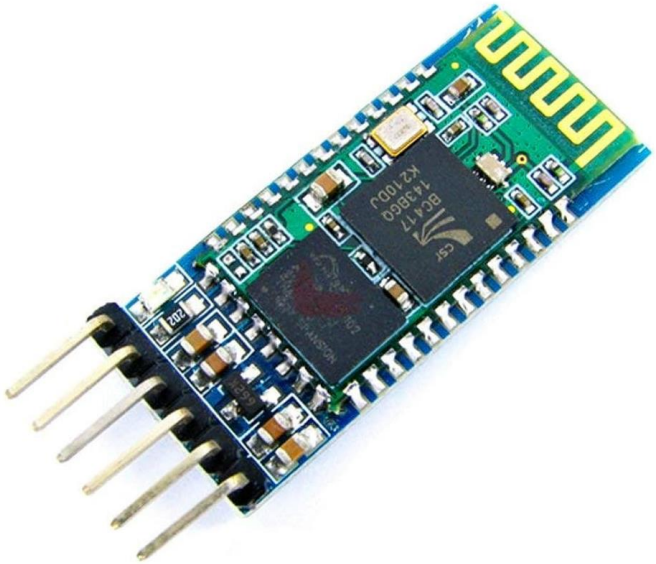
- **Light Intensity Measurement:**
  Measures brightness levels in photography, weather stations, or smart gardens.

- **Solar Tracking Systems:**
  Helps solar panels align towards the brightest sunlight for maximum energy.

- **Smart Home Automation:**
  Adjusts indoor lighting or curtains based on ambient light levels.

Bluetooth Module

A Bluetooth module (e.g., HC-05 or HC-06) is a wireless communication device that allows Arduino to exchange data with smartphones, laptops, or other microcontrollers using Bluetooth technology.



Working Principle:

- Works on **serial communication (UART)**.
- Uses **2.4 GHz radio frequency** to send and receive data wirelessly over short distances (typically 10 meters).
- Arduino sends/receives data via **TX (Transmit)** and **RX (Receive)** pins.

Applications

- **Wireless Control of Robots:**
  Smartphones can send commands via Bluetooth to move robots forward, backward, etc.

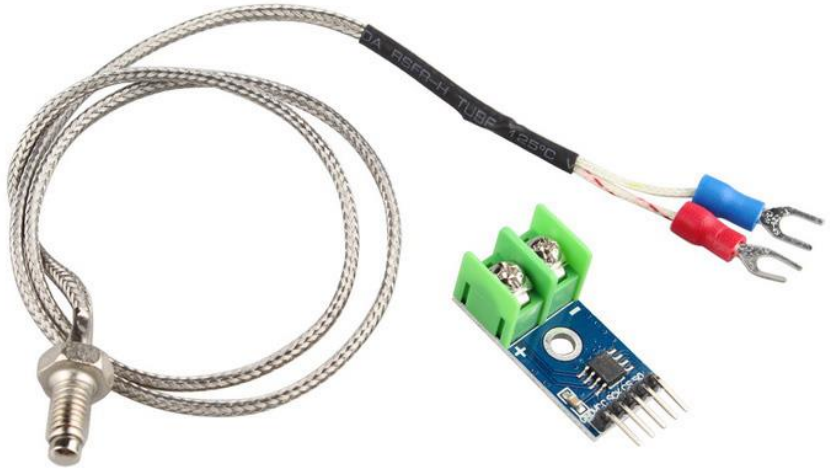
- **Home Automation:**
  Used to control lights, fans, or appliances wirelessly from a mobile app.

- **Data Logging & Monitoring:**
  - *Sends sensor data (temperature, humidity, etc.) to a smartphone for real-time monitoring.*

- **Wireless Communication Between Microcontrollers:**
  Two Arduino boards can exchange data without physical wiring.

Thermocouple Sensor Module

A thermocouple sensor module measures high temperatures by using a thermocouple (two different metal wires joined at one end) and a signal-conditioning module (like MAX6675). It is ideal for industrial or scientific temperature measurements.



Working Principle:

- Based on the **Seebeck Effect**:
  - When two dissimilar metals are joined and exposed to heat, a voltage is generated proportional to the temperature difference between the hot junction (measuring point) and cold junction (reference point).

- The module (MAX6675) converts this small voltage into a digital temperature reading, which the Arduino can read.

Applications

- **Industrial Temperature Monitoring:**
  Measures very high temperatures (up to 800°C or more) in furnaces, kilns, or ovens.

- **3D Printers & Soldering Stations:**
  *Ensures **precise temperature control** of heated nozzles or soldering tips.*

- **Scientific Experiments:**
  Used in laboratories to record accurate high-temperature data.

- **Food & Beverage Industry:**
  Monitors cooking or processing temperatures in commercial kitchens.

References

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